CMPG 322 - Decision Support Systems (DSS) 2
SU3 - Forecasting Theory



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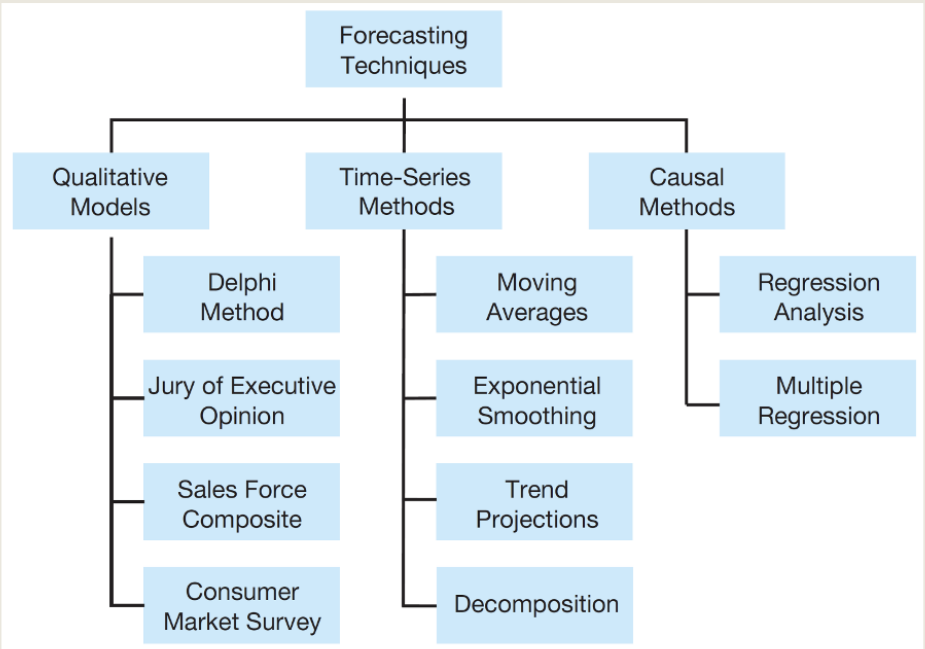
Outline

- Various families of forecasting models;
- Compare moving averages, exponential smoothing, and other time-series models;
- Measures of forecast accuracy;
- Forecast models for random variations;
- Forecast models for trends and random variations;
- How to monitor and control forecasts.

- Main purpose of forecasting:
  - Reduce uncertainty and make better estimates of what will happen in the future
- Subjective methods
  - Seat-of-the pants methods, intuition, experience
- More formal quantitative and qualitative techniques

Forecasting Models

FIGURE 5.1 Forecasting Models



Incorporate judgmental or subjective factors

- Useful when subjective factors are important or accurate quantitative data is difficult to obtain

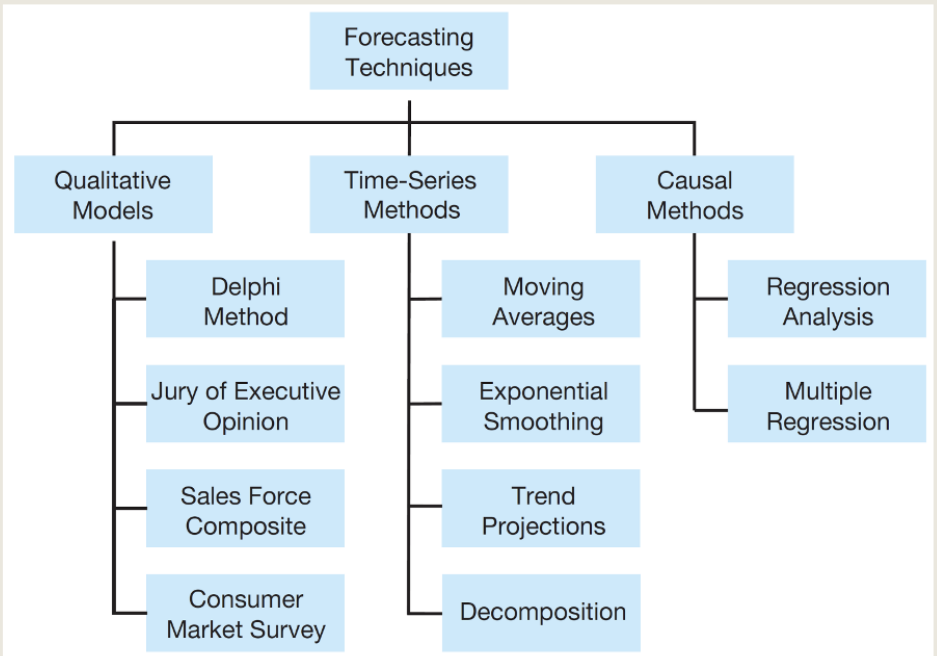




Qualitative Models

Delphi Method

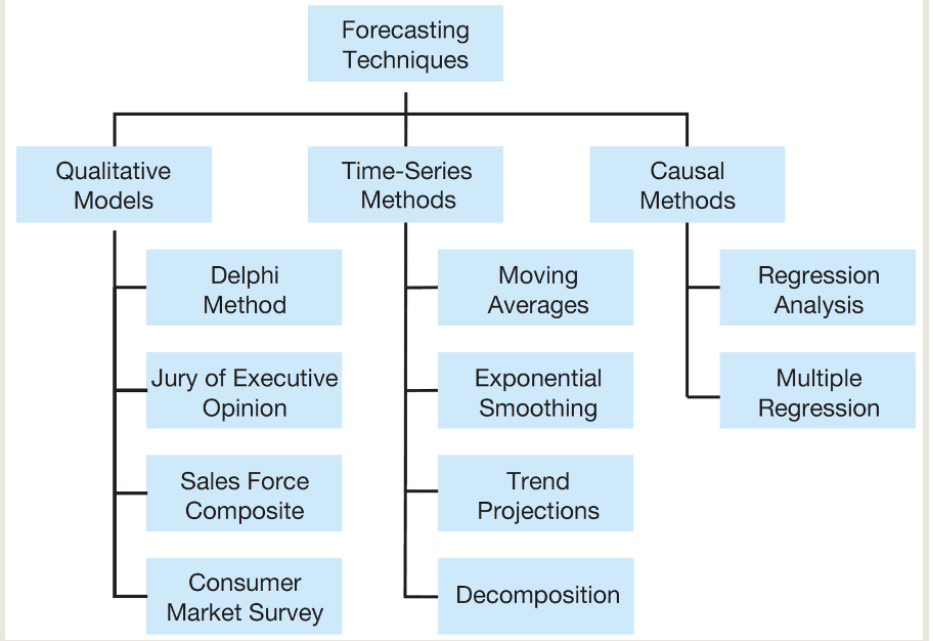
- Iterative group process
- Respondents provide input to decision makers
- Repeated until consensus is reached



Qualitative Models

Jury of Executive Opinion

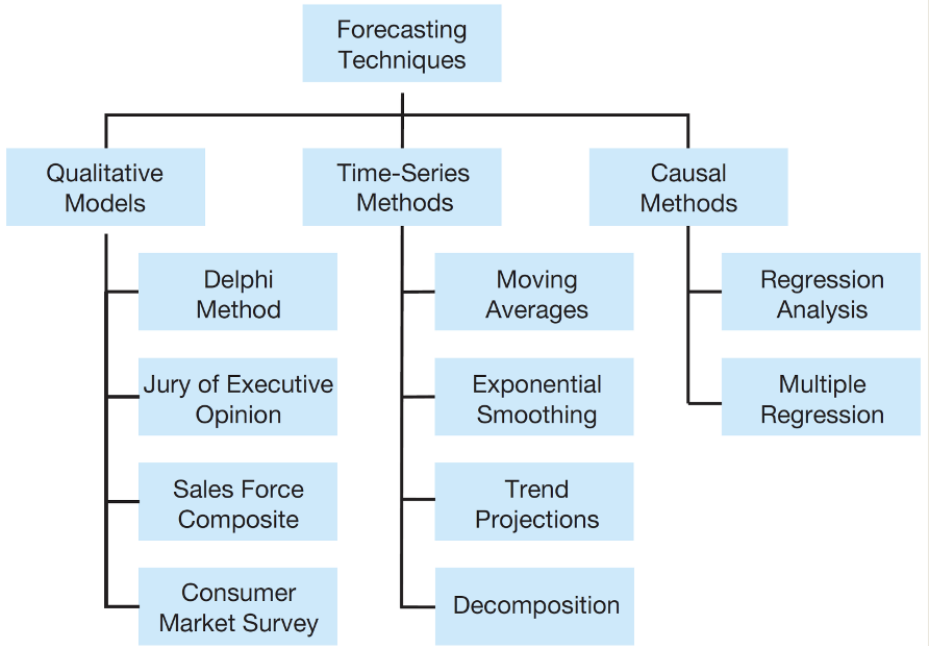
- Collects opinions of a small group of high-level managers
- May use statistical models for analysis



Qualitative Models

Sales Force Composite

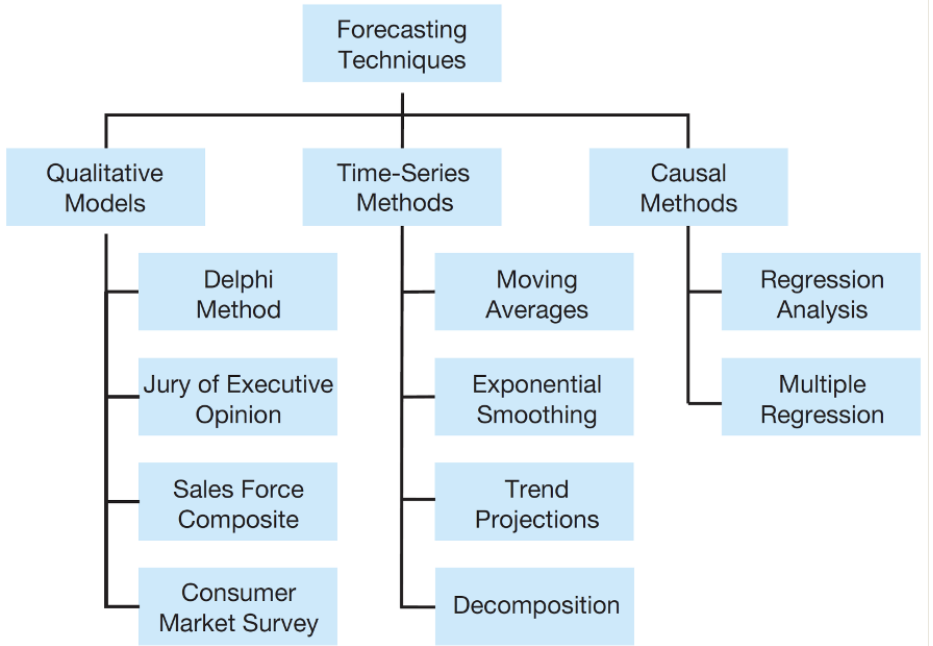
- Allows individual salespersons estimates
- Reviewed for reasonableness
- Data is compiled at a district or national level



Qualitative Models

Consumer Market Survey

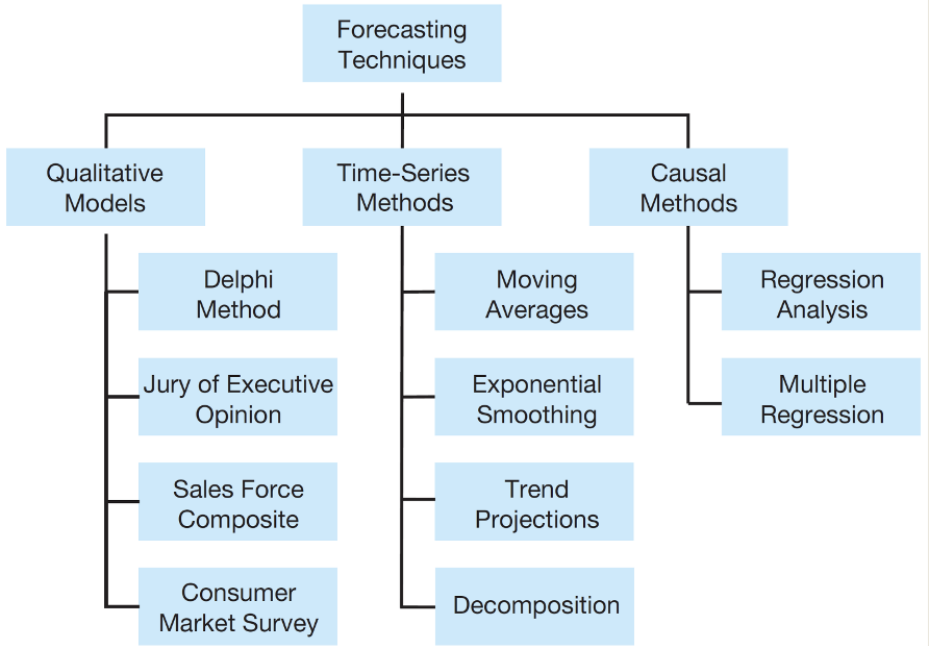
- Information on purchasing plans solicited from customers or potential customers
- Used in forecasting, product design, new product planning



Causal Models

Regression models

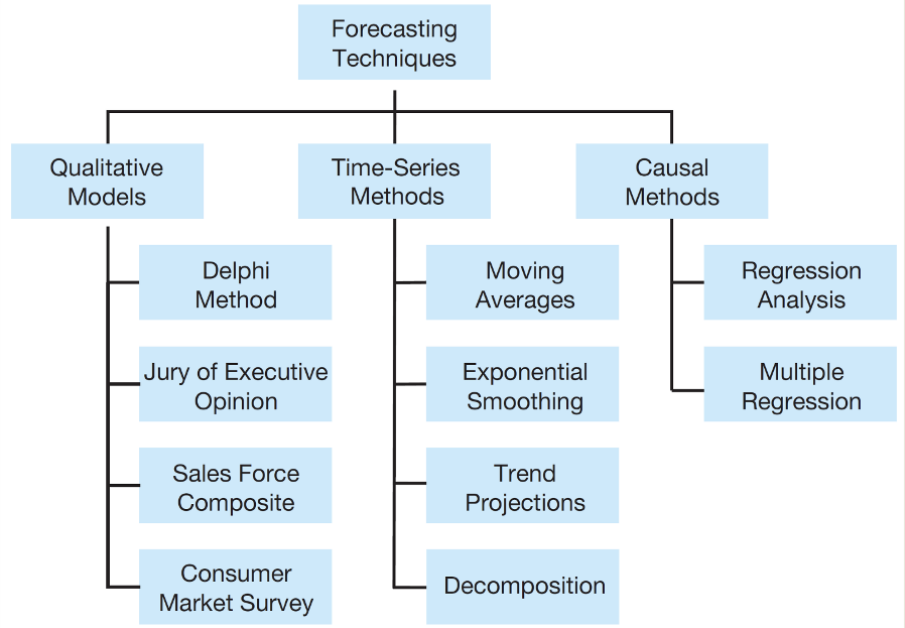
- Causal models if the variable to be forecast is influenced by other variables in the model
- Regression models



Time-Series Models

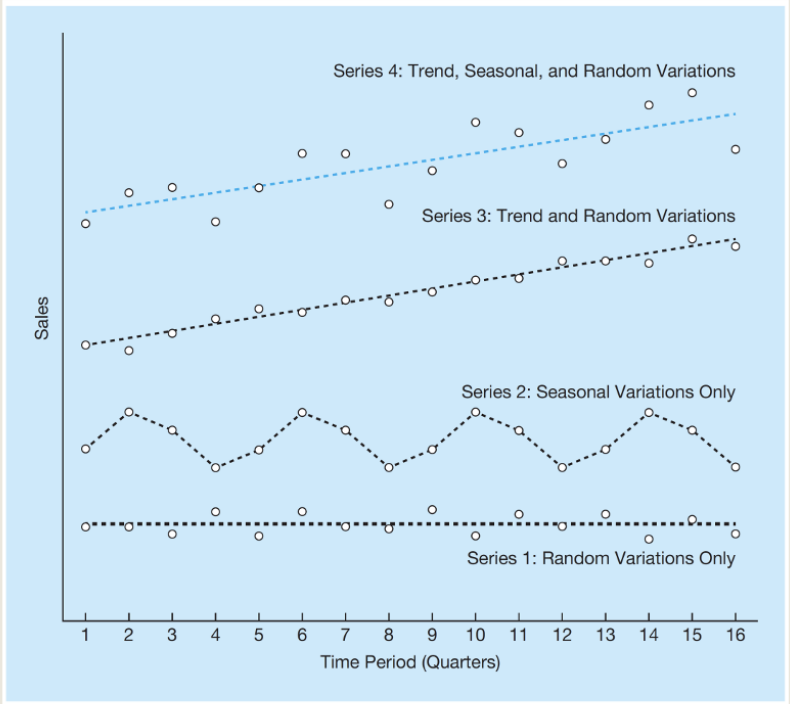
Time-series models

- Predict the future based on the past
- Uses only historical data on one variable
- Extrapolations of past values of a series
- Ignores factors such as economy, competition, selling price



- Sequence of values recorded at successive intervals of time
- Four possible components
  - Trend (T)
  - Seasonal (S)
  - Cyclical (C)
  - Random (R)

FIGURE 5.2 Scatter Diagram for Four Time Series of Quarterly Data





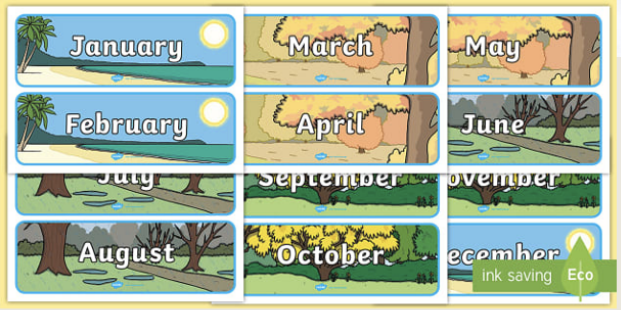






FIGURE 5.3 Scatter Diagram of a Time Series with Cyclical and Random Components

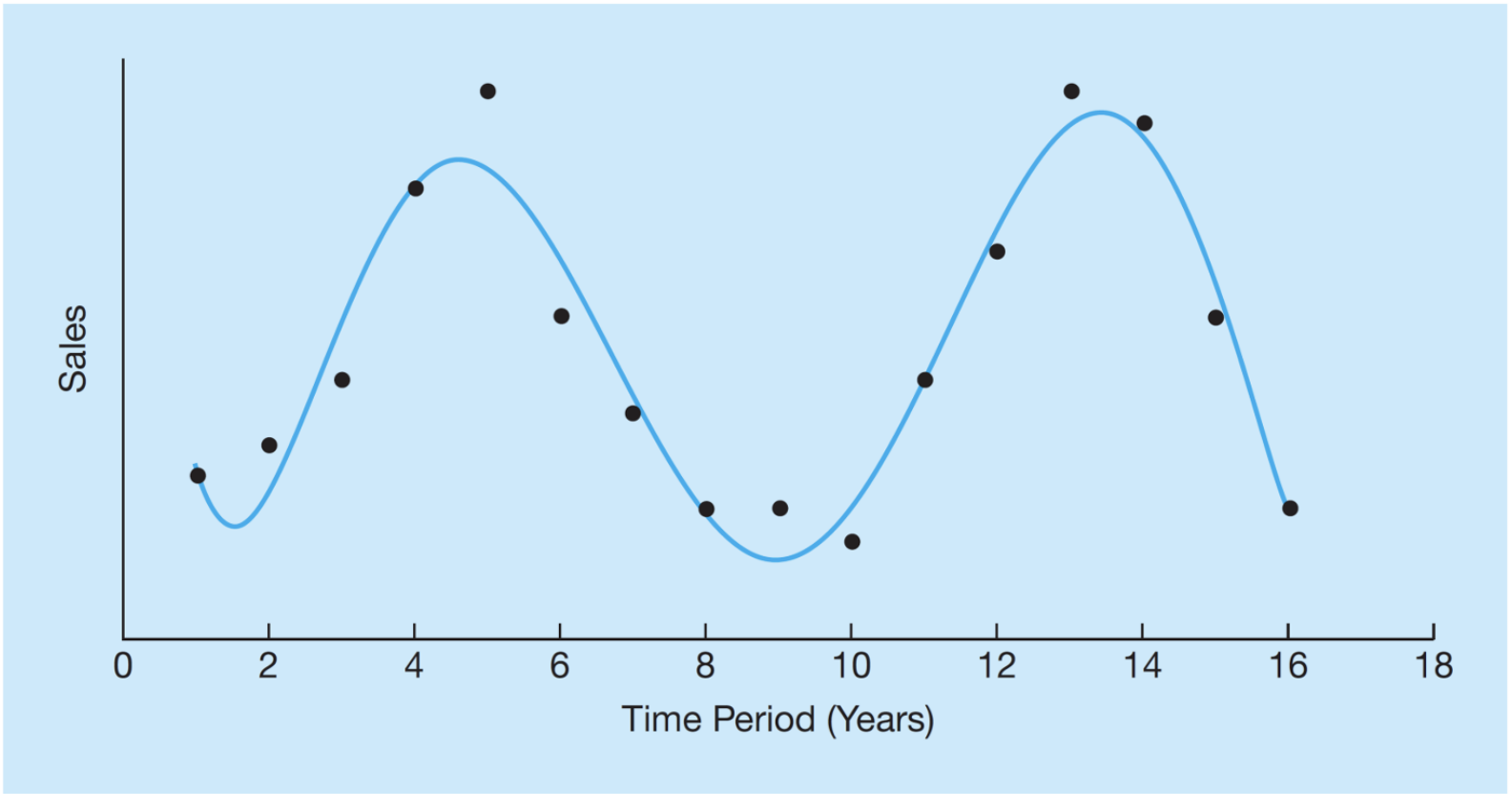
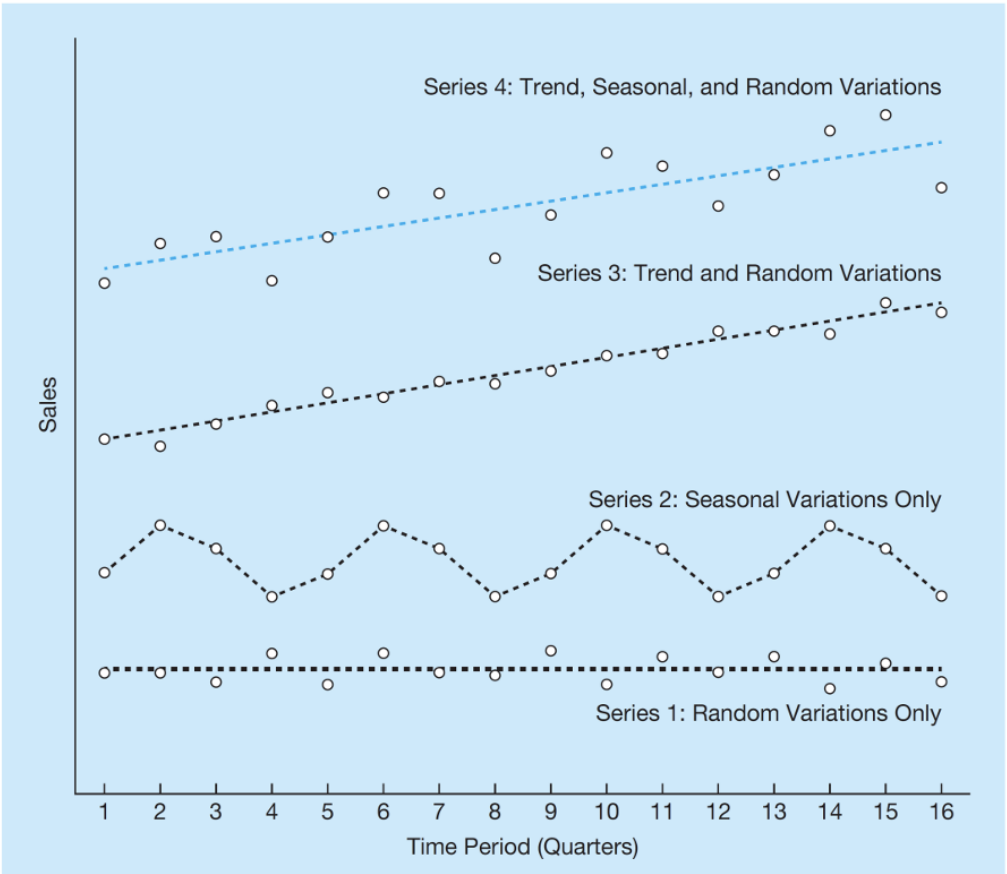


FIGURE 5.2 Scatter Diagram for Four Time Series of Quarterly Data



- **Two basic forms:**
  - Multiplicative
    Demand $= T \times S \times C \times R$
  - Additive
    Demand $= T + S + C + R$
  - Combinations are possible

Measures of forecasting accuracy

- Compare forecasted values with actual values
  - See how well one model works
  - To compare models
- Forecast error = Actual value - Forecast value
- Measure of accuracy
  - Mean absolute deviation (MAD):

$$MAD = \frac{\sum |\text{forecast error}|}{n}$$

TABLE 5.1 Computing the Mean Absolute Deviation (MAD)

Month	Actual Sales of Wireless Speakers	Forecast Sales	Absolute Deviation (Actual - Forecast)
1	110	–	–
2	100	110	$\|100 - 110\| = 10$
3	120	100	$\|120 - 100\| = 20$
4	140	120	$\|140 - 120\| = 20$
5	170	140	$\|170 - 140\| = 30$
6	150	170	$\|150 - 170\| = 20$
7	160	150	$\|160 - 150\| = 10$
8	190	160	$\|190 - 160\| = 30$
9	200	190	$\|200 - 190\| = 10$
10	190	200	$\|190 - 200\| = 10$
11	–	190	–
			Sum of absolute errors = 160 MAD = 160 ÷ 9 = 17.8

Other common measures:

- **Mean squared error (MSE):**

$$MSE = \frac{\sum(\text{error})^2}{n}$$

- **Mean absolute percentage error (MAPE):**

$$MAPE = \frac{\sum \left|\frac{\text{error}}{\text{actual}}\right|}{n} \times 100\%$$

- **Bias** (average error):

$$\text{Bias} = \frac{\sum(\text{error})}{n}$$

- No other components are present
- Averaging techniques smooth out forecasts
  - Moving averages
  - Weighted moving averages
  - Exponential smoothing

- Demand is relatively steady over time
  - Next forecast is the average of the most recent n data values
  - Smooths out short-term irregularities
- Moving average forecast =

$$\frac{\text{Sum of demands in previous } n \text{ periods}}{n}$$

Mathematically

$$F_{t+1} = \frac{Y_t + Y_{t-1} + \ldots + Y_{t-n+1}}{n}$$

where

F_{t+1} = forecast for time period $t+1$

Y_t = actual value in time period t

n = number of periods to average

TABLE 5.2 Wallace Garden Supply Shed Sales

Month	Actual Shed Sales	3-Month Moving Average
January	10	
February	12	
March	13	
April	16	$(10 + 12 + 13) \div 3 = 11.67$
May	19	$(12 + 13 + 16) \div 3 = 13.67$
June	23	$(13 + 16 + 19) \div 3 = 16.00$
July	26	$(16 + 19 + 23) \div 3 = 19.33$
August	30	$(19 + 23 + 26) \div 3 = 22.67$
September	28	$(23 + 26 + 30) \div 3 = 26.33$
October	18	$(26 + 30 + 28) \div 3 = 28.00$
November	16	$(30 + 28 + 18) \div 3 = 25.33$
December	14	$(28 + 18 + 16) \div 3 = 20.67$
January	–	$(18 + 16 + 14) \div 3 = 16.00$

Use weights to put more emphasis on previous periods
- *Often used when a trend or other pattern is emerging*

$$F_{t+1} = \frac{\sum(\text{Weight in period } i)(\text{Actual value in period})}{\sum(\text{Weights})}$$

where

$$F_{t+1} = \frac{w_1 Y_t + w_2 Y_{t-1} + \ldots + w_n Y_{t-n+1}}{w_1 + w_2 + \ldots + w_n}$$

$$w_i = \text{weight for the } i^{th} \text{ observation}$$

Weight Applied	**Period**
$3 \times$ Sales last month	Last month
$2 \times$ Sales 2 months ago	2 months ago
$1 \times$ Sales 3 months ago	3 months ago
6	Sum of the weights

Geweegde bewegende gemiddelde
Weighted moving averages

TABLE 5.3 Weighted Moving Average Forecast for Wallace Garden Supply

MONTH	ACTUAL SHED SALES	3-MONTH WEIGHTED MOVING AVERAGE
January	10	
February	12	
March	13	
April	16	[(3 × 13) + (2 × 12) + (10)]÷6 = 12.17
May	19	[(3 × 16) + (2 × 13) + (12)]÷6 = 14.33
June	23	[(3 × 19) + (2 × 16) + (13)]÷6 = 17.00
July	26	[(3 × 23) + (2 × 19) + (16)]÷6 = 20.5
August	30	[(3 × 26) + (2 × 23) + (19)]÷6 = 23.83
September	28	[(3 × 30) + (2 × 26) + (23)]÷6 = 27.5
October	18	[(3 × 28) + (2 × 30) + (26)]÷6 = 28.33
November	16	[(3 × 18) + (2 × 28) + (30)]÷6 = 23.33
December	14	[(3 × 16) + (2 × 18) + (28)]÷6 = 18.67
January	–	[(3 × 14) + (2 × 16) + (18)]÷6 = 15.33

- Type of moving average
- Requires little record keeping of data

New forecast = Last period's forecast
+ α (Last period's actual demand - Last period's forecast)

α is a weight (or **smoothing constant**) with a value $0 \leq \alpha \leq 1$

$$F_{t+1} = F_t + \alpha(Y_t - F_t)$$

where

F_{t+1} = new forecast (for time period $t+1$)
F_t = previous forecast (for time period t)
α = smoothing constant ($0 \leq \alpha \leq 1$)
Y_t = previous period's actual demand

The idea is simple – the new estimate is the old estimate plus some fraction of the error in the last period

- In January, February's demand for a certain car model was predicted to be 142
- Actual February demand was 153 autos
- Using a smoothing constant of $\alpha = 0.20$, what is the forecast for March?

  New forecast (for March demand) = 142 + 0.2(153 - 142)
  = 144.2 or 144 autos

- If the actual demand in March is 136:

  New forecast (for April demand) = 144.2 + 0.2(136 - 144.2)
  = 142.6 or 143 autos

- Selecting the appropriate value for α is key to obtaining a good forecast
- The objective is always to generate an accurate forecast
- The general approach is to develop trial forecasts with different values of α and select the α that results in the lowest MAD

- **Key points:**
  - High α values make the forecast more responsive to changes
  - Low α values give more stable, smoother forecasts
  - Typical α values range between 0.1 and 0.5

TABLE 5.4 Port of Baltimore Exponential Smoothing Forecasts for $\alpha = 0.10$ and $\alpha = 0.50$

Quarter	Actual Tonnage Unloaded	Forecast using $\alpha = 0.10$	Forecast using $\alpha = 0.50$
1	180	175	175
2	168	175.5 = 175.00 + 0.10(180 - 175)	177.5
3	159	174.75 = 175.50 + 0.10(168 - 175.50)	172.75
4	175	173.18 = 174.75 + 0.10(159 - 174.75)	165.88
5	190	173.36 = 173.18 + 0.10(175 - 173.18)	170.44
6	205	175.02 = 173.36 + 0.10(190 - 173.36)	180.22
7	180	178.02 = 175.02 + 0.10(205 - 175.02)	192.61
8	182	178.22 = 178.02 + 0.10(180 - 178.02)	186.30
9	?	178.60 = 178.22 + 0.10(182 - 178.22)	184.15

TABLE 5.5 Absolute Deviations and MADs for the Port of Baltimore Example

Quarter	Actual Tonnage	Forecast with $\alpha = 0.10$	Absolute Deviations for $\alpha = 0.10$	Forecast with $\alpha = 0.50$	Absolute Deviations for $\alpha = 0.50$
1	180	175	5	175	5
2	168	175.5	7.5	177.5	9.5
3	159	174.75	15.75	172.75	13.75
4	175	173.18	1.82	165.88	9.12
5	190	173.36	16.64	170.44	19.56
6	205	175.02	29.98	180.22	24.78
7	180	178.02	1.98	192.61	12.61
8	182	178.22	3.78	186.30	4.3

Sum of absolute deviations = 82.45

$$\text{MAD} = \frac{\sum |\text{deviations}|}{n}$$

MAD = 10.31

Best choice

Sum of absolute deviations = 98.54

$$\text{MAD} = \frac{\sum |\text{deviations}|}{n}$$

MAD = 12.32

- Exponential smoothing does not respond to trends
- The basic approach:
  1. Develop an exponential smoothing forecast
  2. Adjust it for the trend

Revise exponential smoothing

$$F_{t+1} = F_t + \alpha(Y_t - F_t)$$

where

F_{t+1} = new forecast (for time period $t+1$)

F_t = previous forecast (for time period t)

α = smoothing constant ($0 \leq \alpha \leq 1$)

Y_t = previous period's actual demand

$$F_{t+1} = FIT_t + \alpha(Y_t - FIT_t)$$
$$T_{t+1} = T_t + \beta(F_{t+1} - FIT_t)$$
$$FIT_{t+1} = F_{t+1} + T_{t+1}$$

- The equation for the trend correction uses a new smoothing constant β
- F_t and T_t must be given or estimated

T_t = smoothed trend for time period t
F_t = smoothed forecast for time period t
FIT_t = forecast including trend for time period t
α = smoothing constant for forecasts
β = smoothing constant for trend

- Step 1: Compute smoothed forecast F_{t+1}
  Smoothed forecast = Previous forecast including trend + α(Last error)
  $F_{t+1} = FIT_t + \alpha(Y - FIT_t)$
- Step 2: Update the trend (T_{t+1}) using
  Smoothed trend = Previous forecast including trend + β(Error or excess in trend)
  $T_{t+1} = T_t + \beta(F_{t+1} - FIT_t)$
- Step 3: Calculate the trend-adjusted exponential smoothing forecast (FIT_{t+1}) using
  Forecast including trend (FIT_{t+1}) = Smoothed forecast (F_{t+1}) + Smoothed trend (T_{t+1})
  $FIT_{t+1} = F_{t+1} + T_{t+1}$

- A high value of β makes the forecast more responsive to changes in the trend
- A low value of β gives less weight to the recent trend and tends to smooth out the trend
- Values are often selected using a trial-and-error approach based on the value of the MAD for different values of β

Example: Midwestern Manufacturing

YEAR	ELECTRICAL GENERATORS SOLD
2007	74
2008	79
2009	80
2010	90
2011	105
2012	142
2013	122

TABLE 5.6 Midwestern Manufacturing's Demand

For 2009 (time period 3)

Step 1: $F_3 = FIT_2 + \alpha(Y_2 - FIT_2)$

$= 74 + 0.3(79 - 74) = 75.5$

Step 2: $T_3 = T_2 + .4(F_3 - FIT_2)$

$= 0 + .4(75.5 - 74) = 0.6$

Step 3: $FIT_3 = F_3 + T_3$

$= 75.5 + 0.6 = 76.1$

Example: Midwestern Manufacturing

	A	B	C	D	E	F	G	H	I	J
1	**Midwestern Manufacturing Company Example**									
2										
3	**Forecasting**		**Trend adjusted exponential smoothing**							
4	Enter alpha and beta (between 0 and 1), enter the past demands in the shaded column then enter a starting forecast. If the starting forecast is not in the first period then delete the error analysis for									
5										
6										
7	Alpha	0.3								
8	Beta	0.4								
9	Data			**Forecasts and Error Analysis**						
10	Period	Demand		Smoothed Forecast, F_t	Smoothed Trend, T_t	Forecast Including Trend, FIT_t	Error	Absolute	Squared	Abs Pct Err
11	Period 1	74		74		74	0	0	0	00.00%
12	Period 2	79		74	0	74	5	5	25	06.33%
13	Period 3	80		75.5	0.6	76.1	3.9	3.9	15.21	04.88%
14	Period 4	90		77.27	1.068	78.338	11.662	11.662	136.0022	12.96%
15	Period 5	105		81.8366	2.46744	84.30404	20.69596	20.696	428.3228	19.71%
16	Period 6	142		90.512828	4.950955	95.4637832	46.53622	46.5362	2165.619	32.77%
17	Period 7	122		109.4246482	10.5353	119.959949	2.040051	2.04005	4.161806	0.016722
18		**Next period**		**120.5719646**	**10.78011**	**131.352072**				
19				Total			89.83423	89.8342	2774.316	78.32%
20							**12.83346**	**12.8335**	**396.3309**	**11.19%**
21							**Bias**	**MAD**	**MSE**	**MAPE**
22								**SE**	**23.55554**	

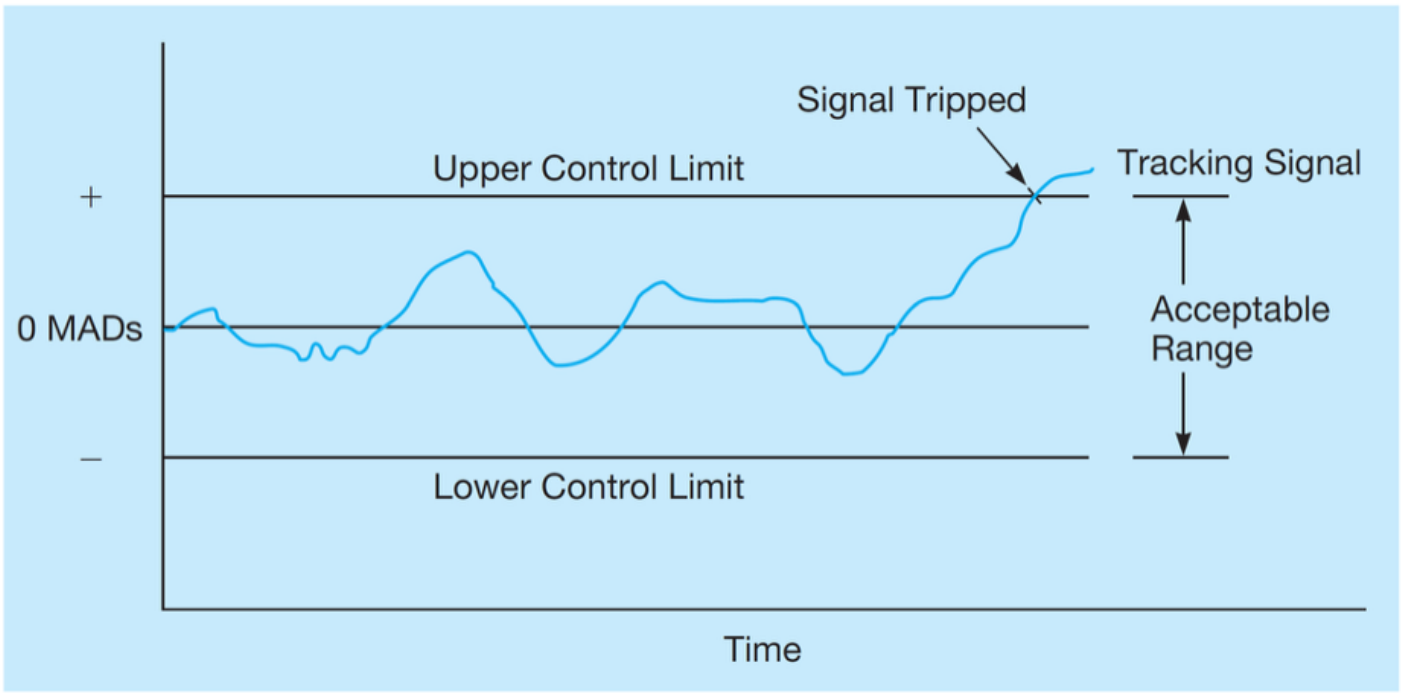
The forecast for next period is here.

- Tracking signal measures how well a forecast predicts actual values
- Running sum of forecast errors (RSFE) divided by the MAD

$$\text{Tracking signal} = \frac{\text{RSFE}}{\text{MAD}} = \frac{\sum(\text{forecast error})}{\text{MAD}}$$

$$\text{MAD} = \frac{\sum|\text{forecast error}|}{n}$$

FIGURE 5.7 Plot of Tracking Signals



Kimball's Bakery

Quarterly sales of croissants (in thousands)

TIME PERIOD	FORECAST DEMAND	ACTUAL DEMAND	ERROR	RSFE	|FORECAST| | ERROR |	CUMULATIVE ERROR	MAD	TRACKING SIGNAL
1	100	90	−10	−10	10	10	10.0	−1
2	100	95	−5	−15	5	15	7.5	−2
3	100	115	+15	0	15	30	10.0	0
4	110	100	−10	−10	10	40	10.0	−1
5	110	125	+15	+5	15	55	11.0	+0.5
6	110	140	+30	+35	35	85	14.2	+2.5

For Period 6:

$$\text{MAD} = \frac{\sum |\text{forecast error}|}{n} = \frac{85}{6} = 14.2$$

$$\text{Tracking signal} = \frac{\text{RSFE}}{\text{MAD}} = \frac{35}{14.2} = 2.5 \text{ MADs}$$

Questions

